

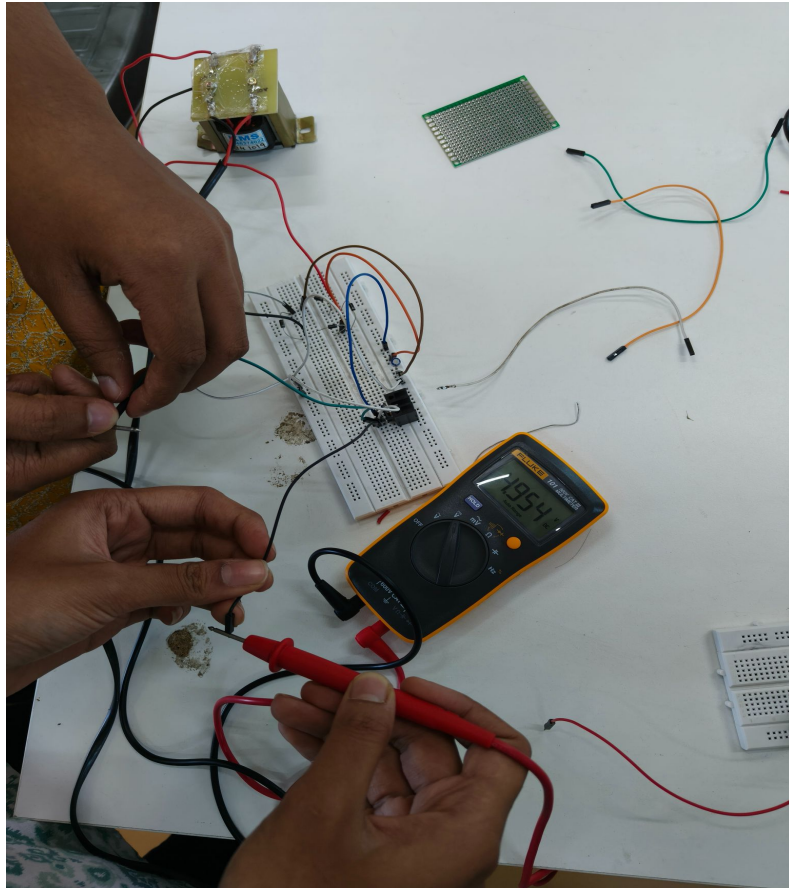
PROGRESS LOG

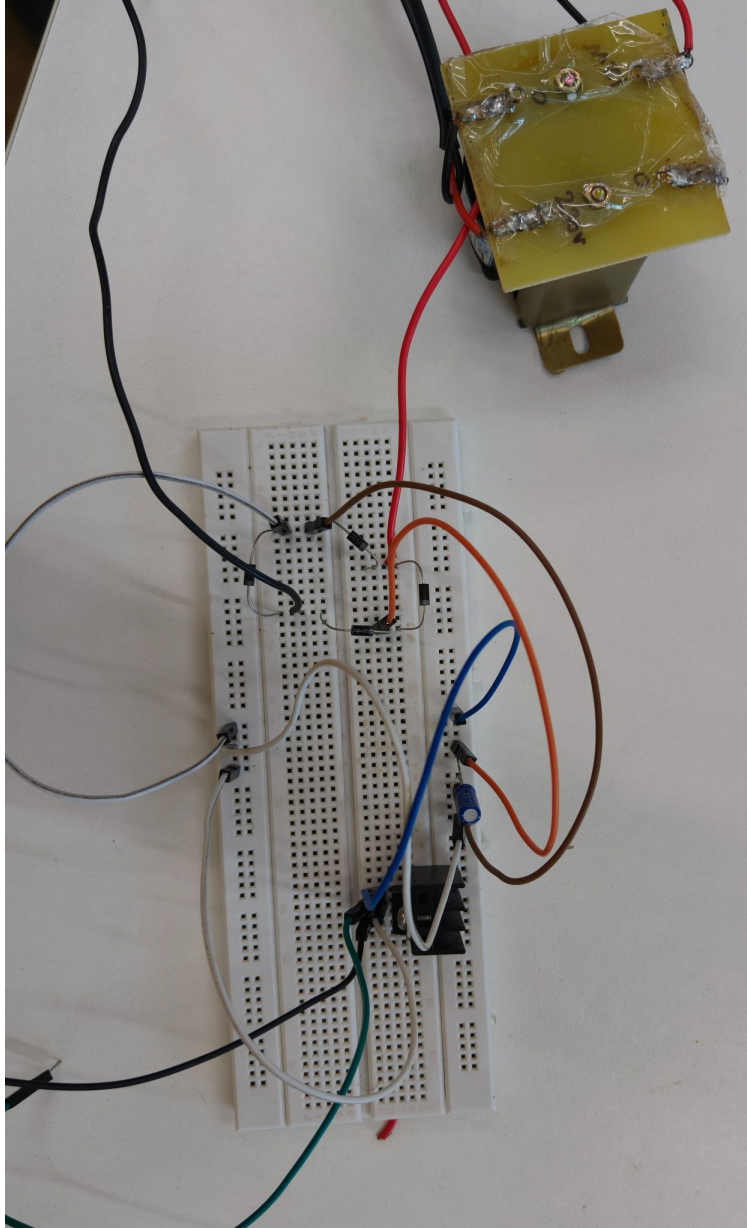
-Bhavajna,Lasya

All the circuits were first built on a breadboard. Only after choosing the final circuit, we used the perfboard and soldered the components.

The First Circuit We Built

We initially tried building the circuit available in Arattai, and after testing it, we found that it was working successfully. It showed us 5v on multimeter





Research On The Constructed Circuit And Its Modifications

Sources used for research

Gemini

<https://youtu.be/ZMys3FkI9RM> (for capacitor filter)

https://youtu.be/FE_peKaYGF4(for basics of rectifier)

Chatgpt

During our research on this circuit, we came across the concept of heat dissipation in components such as the 7805 voltage regulator, diodes, and transformer. We noticed that excessive heat can affect the performance and efficiency of these components. We thought we may need an external heat sink but we got to know that the 7805 used in our lab setup is equipped with a heat sink to help dissipate heat.

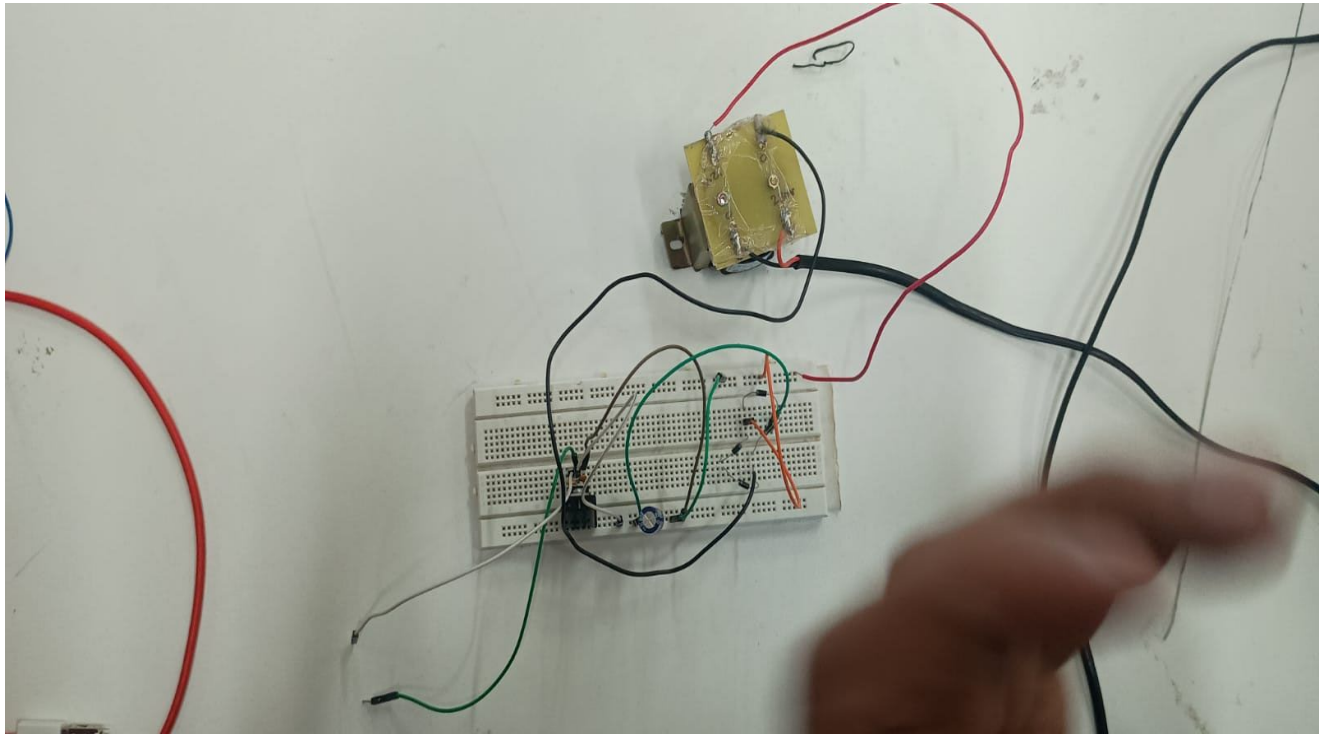
We further learned that excessive ripple voltage at the input of the 7805 can affect its proper functioning, as the regulator requires a minimum input voltage of approximately 7.5–8 V to maintain a stable 5 V output. Since a capacitor with a higher capacitance provides better smoothing of the rectified output and reduces ripple voltage, we considered replacing the existing 100 μF capacitor with a 1000 μF capacitor. We believed(after watching some youtube videos) this would be a more effective way to reduce ripple and improve the stability of the 7805, rather than increasing the transformer's output voltage, which could result in greater power dissipation and additional heating in the regulator.

During our research, we found that high-frequency ripples and noise can also occur in the circuit. Therefore, we added a

0.1 μF capacitor near the 7805, which provides a low-impedance path to ground for high-frequency noise, supplies instantaneous energy during sudden load changes, and helps prevent internal oscillations, improving the regulator's stability.

A small error

We initially intended to build the modified circuit on a breadboard. However, we accidentally reversed the direction of one of the diodes and used one of the AC output wires as the common ground rail. As a result, the circuit did not work as expected. We then corrected the circuit connections, rebuilt it properly on the breadboard, and tested it again. The circuit worked successfully after making the necessary corrections.



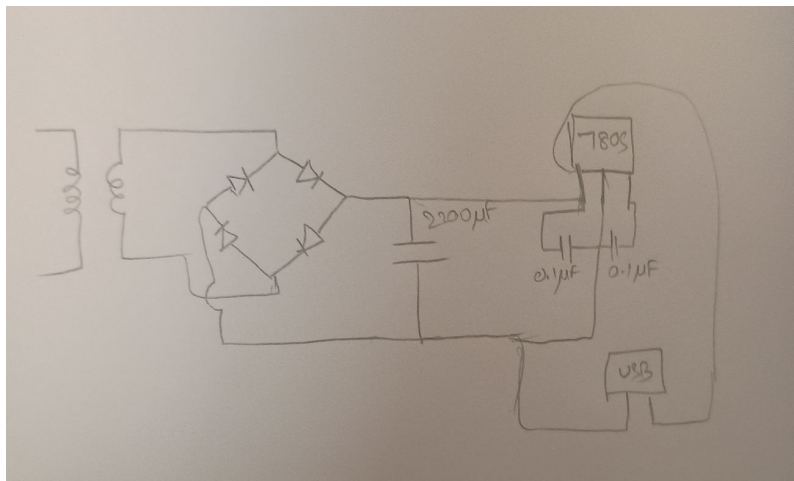
The Intermediate Circuit

We corrected the circuit, tried it on the breadboard and then soldered the components onto a perfboard.

During soldering, we noticed that the wires and the perfboard became heated due to prolonged contact with the soldering iron (as copper has high thermal conductivity) during soldering.

We had to continuously pause during soldering so that the wires or the perfboard don't burn up. After completing the assembly, the circuit worked successfully.

Circuit diagram:



Our intermediate model:

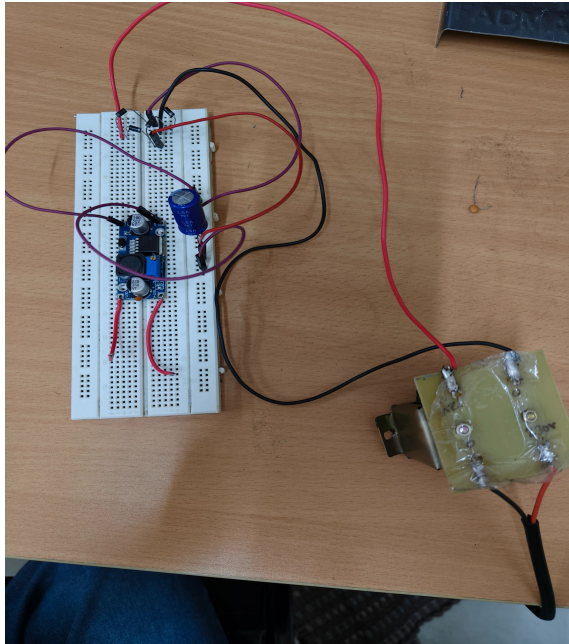
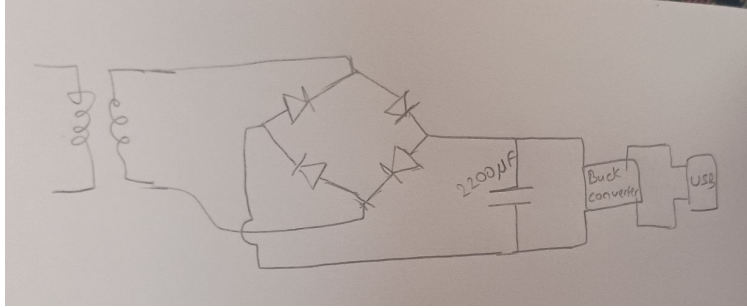


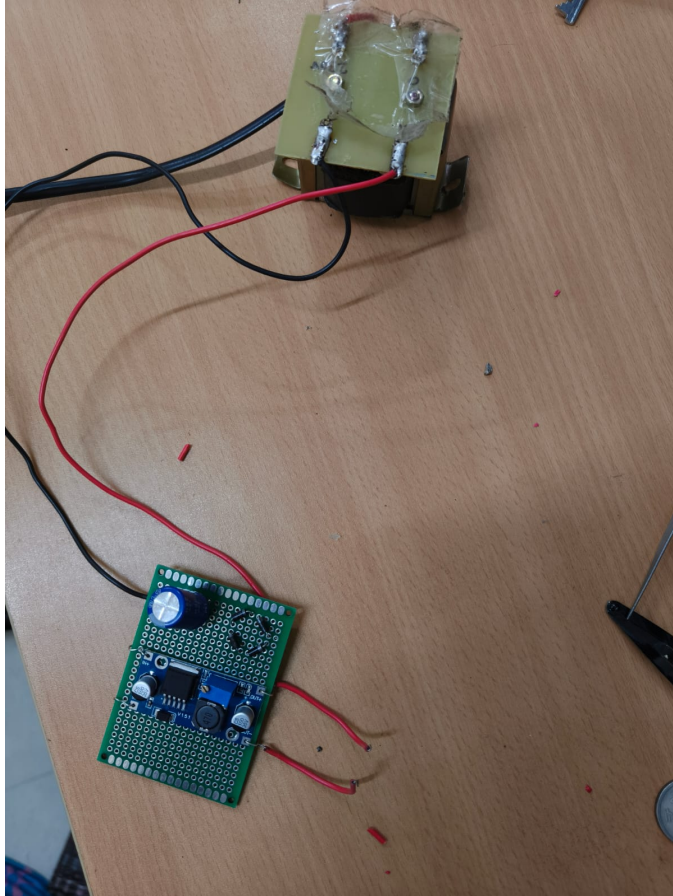
Trials for a smart charger

Then we found that the circuit was not providing a stable power supply to the phone, possibly due to overheating and voltage ripples. We wanted to make a smart charger that can regulate the power supply based on the battery percentage (which can be recognised by the voltage difference and the current the phone is taking) for which we would use an atmega chip. As the connections to the chip are very complicated, we thought of using an arduino instead. Arduino however can handle current in mA so the rest of the current had to be diverted using a mosfet. Due to the high amount of current through the mosfet, we needed to use a heat sink which wasn't available so we had to drop the idea.

Final circuit

We later decided to replace the existing regulator with a buck converter. We also added a 2200 μF capacitor to reduce low-frequency noise and improve the stability of the output voltage. The following circuit shows the modified setup implemented on the breadboard.





3D printing

We started designing the 3D models after the final circuit was done. We started learning freecad once the circuit using the 7805 was done. After the deadline was extended, we decided to switch to openscad as it was coding and less time taking. We took the help of AI to write the code and learnt the idea behind how the code and openscad works.

3D model contains :

1. The container to place the transformer and the perfboard

2. Two rectangular pieces to cover the opening for the wire from the transformer that connects to the main supply
3. Two rectangular pieces to cover the gap caused by the variation in size of container due to difference in size of the perfboard and transformer.
4. A small cuboid that would be stuck to the wall with the hole for the usb female port to hold it.
5. A sliding lid to close the box, thus making a usb based adaptor.

Our overnight charger

The charger is a slow charging one(by modern standards). It delivers around 5V. Well what is the advantage? My phone's maximum capacity is around 6000mAh, thus it would take around 5.5 to 6.5 hours to charge with this charger. The advantage here is that I can charge it overnight. While normal chargers can also be used for overnight charging, it reduces the battery life very easily. A low power charger helps keep the battery safe while also charging overnight so that you don't have to worry about charging your phone in during a busy day.